🌱 PotGotchi Build Guide

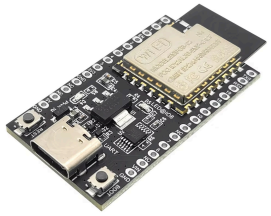
A plant-growing Tamagotchi-style pet built on the ESP32-C3 — from parts to case assembly

Project	**PotGotchi (plant-growing virtual pet)**
About	A small DIY electronic pet made with an ESP32-C3 and a tiny display. Give it water and sunlight and watch it grow from a seed to a flower in 5 stages.
License	GPL-3.0 Distributed under the license of the original project
Changes	Redesigned from a cat pet into a plant pet · Moisture / Sunlight / Growth stats · New Vitality system · New 5-stage growth sprites · Water / love effect animations · Removed unused features (Sleep, Settings)
Credits	Original software: Esp32-Tamagotchi by derdacavga (DSN Industries) (GPL-3.0) Display library: DSN-Esp32C3-St7789 (GPL-3.0) The 3D-printed case (STL) is an independent work, separate from the software.

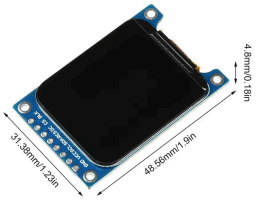
Contents

1. Parts
2. Wiring
3. Display library setup
4. Download and upload the code
5. Assembling the case

1 Parts



① **ESP32-C3 board**



② **1.69" TFT LCD (ST7789V3, 240×280)**

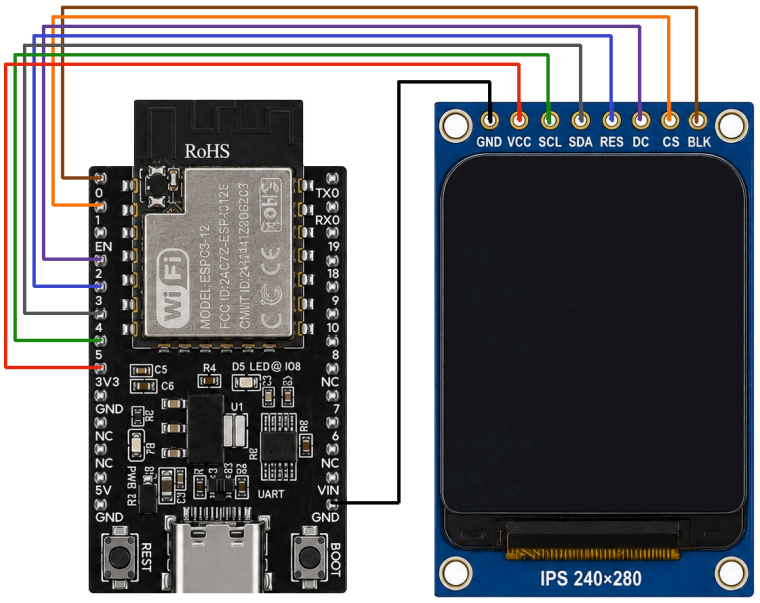


③ **1x4 button module (wires soldered directly)**

Part	Notes
① ESP32-C3 dev board	USB-C, for uploading and power
② 1.69" TFT LCD display	ST7789V3 driver, 240×280, SPI interface
③ 1x4 button module	4 independent buttons. No pin header: strip the insulation from jumper wires and **solder them directly** to the 5 pads. Soldering is not required if you do not use the case.
Jumper wires	8 for the display + wires for the button module. One end of each button wire is stripped and soldered to the module pad.
3D-printed parts	Front case ×1, back plate ×1, board cap ×1, button caps ×4
M3×5 screws + small Phillips screwdriver	To fasten the button module, ×2
USB Type-C cable	For uploading and power (must support data transfer)

2 Wiring

2-1. Board – Display

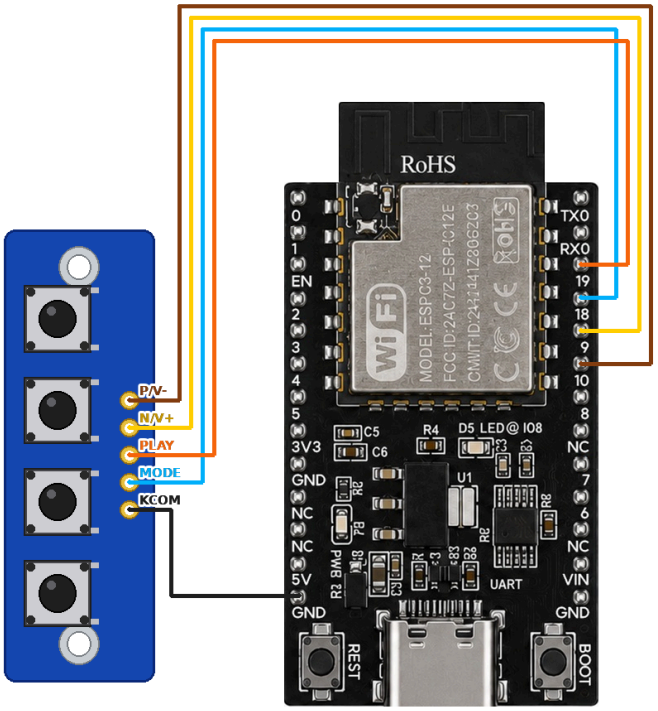


Display pins: GND · VCC · SCL · SDA · RES · DC · CS · BLK

Connect the 8 display pins to the board by following the colored wires in the picture. The table below lists the same connections.

Display pin	Board pin	Wire color
GND	GND (bottom right)	● Black
VCC	3V3	● Red
SCL	**GPIO 5**	● Green
SDA	**GPIO 4**	● Gray
RES	**GPIO 3**	● Blue
DC	**GPIO 2**	● Purple
CS	**GPIO 1**	● Orange
BLK	**GPIO 0**	● Brown

2-2. Board – Button module



1x4 button module and board connections

Module pad	Function	Board pin	Wire color
KCOM (common)	GND	GND (bottom left)	● Black
MODE	Move left (Scroll Left)	**GPIO 18**	● Light blue
PLAY	Select / confirm (OK)	**GPIO 19**	● Orange
N/V+	Move right (Scroll Right)	**GPIO 9**	● Yellow
P/V-	Give love (Love)	**GPIO 10**	● Brown

> Each button connects to KCOM (GND) when pressed, so no resistor is needed. Connect them directly. The pin numbers in the code are defined in `HardwareConfig.h`.

3 Display library setup

3-1. Download the library

This project uses the display library by derdacavga.

```
https://github.com/derdacavga/DSN-Esp32C3-St7789
```

1. Download the library from the repository above.
2. Install it in the Arduino IDE by following the instructions there.

3-2. Edit User_Setup.h — the most important step

After installing the library, find the `User_Setup.h` file in this path:

```
Documents\Arduino\libraries\DSN_Esp32C3_St7789\src\User_Setup.h
```

Open the file and change the values below. (They are in the "DISPLAY RESOLUTION" and "HARDWARE PINS" sections of the file.)

Item	Before	After
`TFT_HEIGHT`	240	**280**
`TFT_CS`	-1	**1**
`TFT_DC`	8	**2**
`TFT_RST`	5	**3**
`TFT_MOSI`	6	**4**
`TFT_SCLK`	4	**5**
`TFT_BL`	2	**0**

After editing, it should look like this:

```
// ========================================================
// 1. DISPLAY RESOLUTION
// ========================================================
#define TFT_WIDTH  240
#define TFT_HEIGHT 280

// ========================================================
// 2. HARDWARE PINS (ESP32-C3)
// ========================================================
#define TFT_MISO -1  // set to -1 if not connected, or if shared with SD card connect to 1
#define TFT_CS   1   // Chip Select
#define TFT_DC   2   // Data/Command
#define TFT_RST  3   // Reset (Set to -1 if tied to 3.3V)
#define TFT_MOSI 4   // SPI Data
#define TFT_SCLK 5   // SPI Clock
#define TFT_BL   0   // Backlight (Optional, set to -1 if not used)
```

After editing

> **If you skip this step**, the screen resolution may be wrong or the display may not respond at all. The library defaults assume a different panel size (240×240) and a different pin layout, so these values must be changed to match the 1.69" (240×280) panel and the wiring above. The pin numbers must **match the wiring in chapter 2**.

4 Download and upload the code

4-1. Download the code

```
https://github.com/hane94/PotGotchi
```

1. Open the repository above, click the green **Code** button → **Download ZIP**.
2. Unzip it and open `PotGotchi.ino` in the Arduino IDE.

> **Important:** All `.h` and `.cpp` files must be in the **same folder** as `PotGotchi.ino` for it to compile.

4-2. Board settings and upload

1. Connect the board to your computer with a USB cable.
2. In the Arduino IDE, select the board **ESP32C3 Dev Module**.
3. Click the upload button to upload the code.
4. When the upload finishes, check that the plant picture appears on the screen.

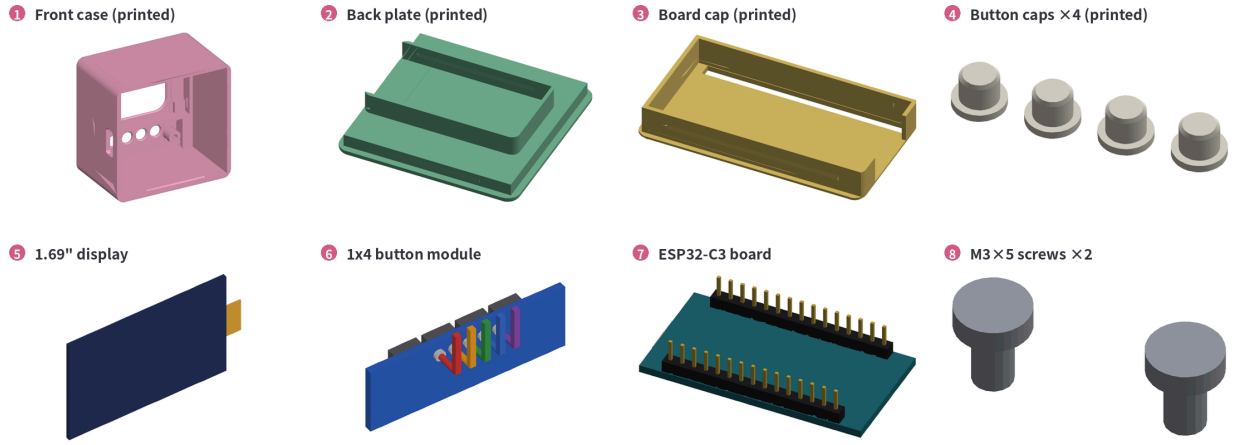
5 Assembling the case

Download the case

Download	Get the 3D case models (STL) from MakerWorld. **Link:** https://makerworld.com/en/models/3476394-potgotchi-diy-mini-game-plant-tamagotchi
Files	Front case, back plate, board cap, button caps ×4

Before assembling: Finish chapters 1–4 and check that the screen and buttons work *before* putting anything in the case. Fixing problems after assembly means taking everything apart again.

5-1. Parts to assemble

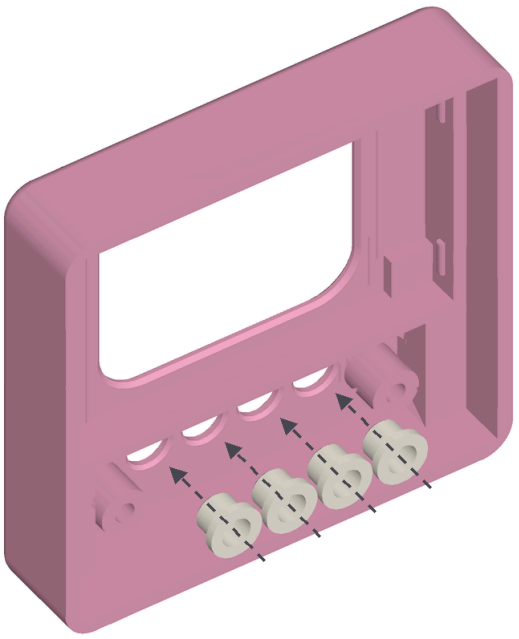


4 printed parts (①–④) and the electronics and screws (⑤–⑧)

Additional items: Printed parts — front case, back plate, board cap, 4 button caps. Screws — 2 × M3×5 and a small Phillips screwdriver.

5-2. Front case — insert the button caps

❶ **Button caps ×4 — insert from the inside into the holes**

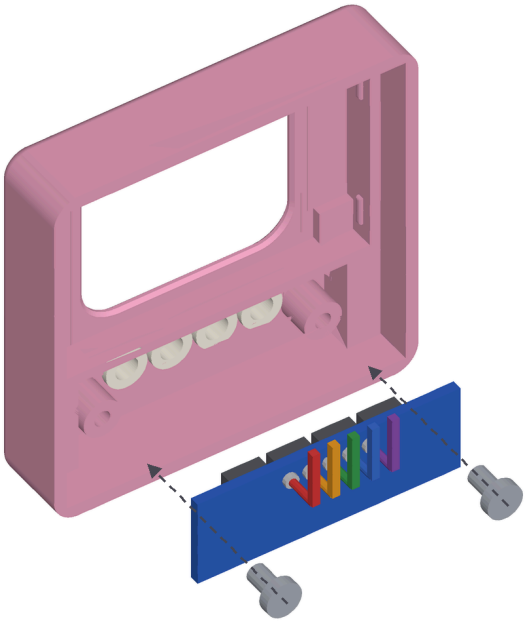


The case is shown cut in half to make the structure easier to see

1. Turn the front case over so the screen window faces down. (The inside faces up.)
2. Insert each button cap into the 4 button holes **from the inside**. The wide plate (flange) stays on the inside and the thin stem comes out through the hole.
3. From the front, the caps should move slightly inside the holes. That is normal.

5-3. Front case — fasten the button module

❷ **Button module — switches facing the front (cap side)**
❸ **M3×5 screws ×2 — fasten into the screw posts**



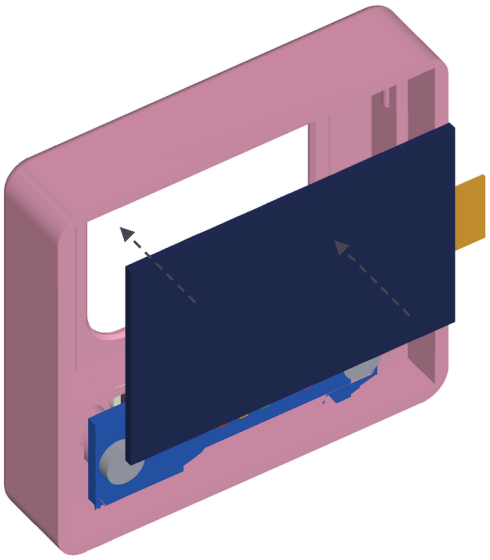
1. Face the switches of the button module toward the front (cap side) and line up its two mounting holes with the screw posts inside the case.

2. Fasten it with 2 × M3×5 screws. The screws cut their own thread into the post. **Do not overtighten**; stop when the board touches the posts.
3. Press the caps and check that each button clicks.

> The button module has **jumper wires soldered directly to the pads**, with no pin header. A pin header would stick out toward the front (cap side) and make assembly difficult, so the wires are soldered to come out toward the back (board side).

5-4. Front case — insert the display

❹ **Display — screen facing the window**

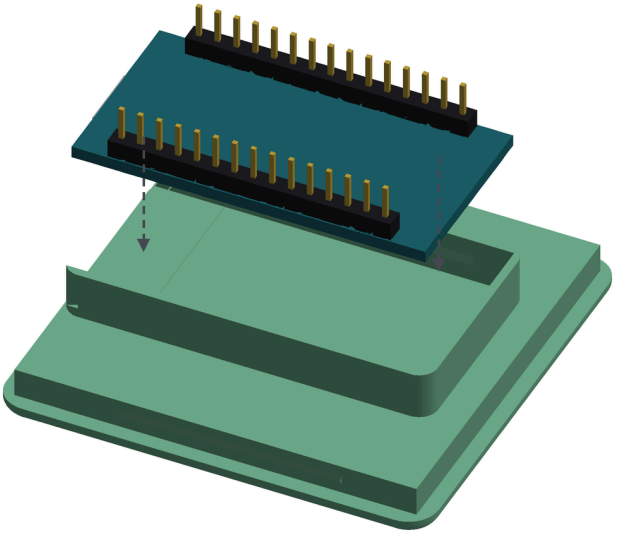


1. Place the display in the display seat inside the front case with the glass facing the screen window.
2. Do not glue the display. The pressing posts of the back plate will hold it down at the corners later.

> Do not press on the parts on the back of the display (IC, capacitors) or on the orange FPC. Do not press hard anywhere except the corners.

5-5. Back plate — install the dev board first

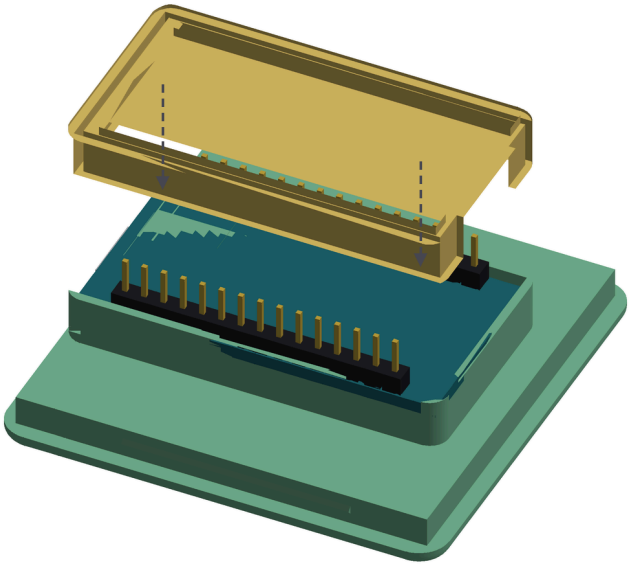
❶ **ESP32-C3 board — USB-C at the left end, pin headers facing up**
❷ **Placed in the cradle of the back plate**



1. Place the back plate on the table. You will see the raised cradle.

2. Set the dev board in the cradle with the **USB-C at the left end** and the pin headers facing up.

❸ Board cap — the 2 slots sit over the pin headers

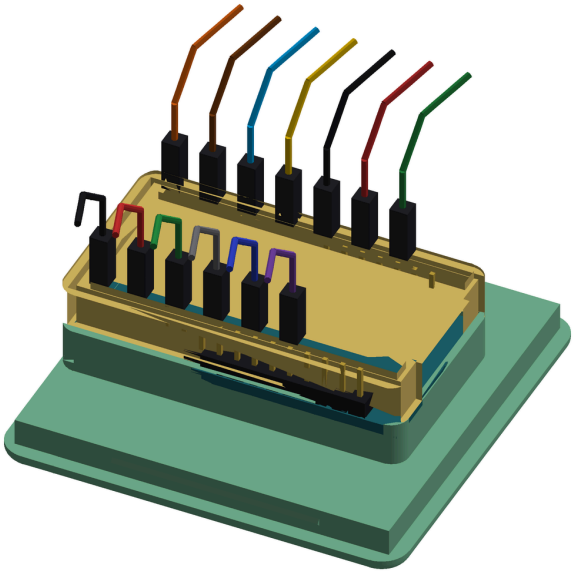


3. Put the board cap on from above. The 2 slots of the cap must sit over the rows of pin headers.

> **The order matters:** Secure the dev board in the back plate first, then plug in the jumper wires. If the jumper wires are plugged in first, the board will not fit into the cradle.

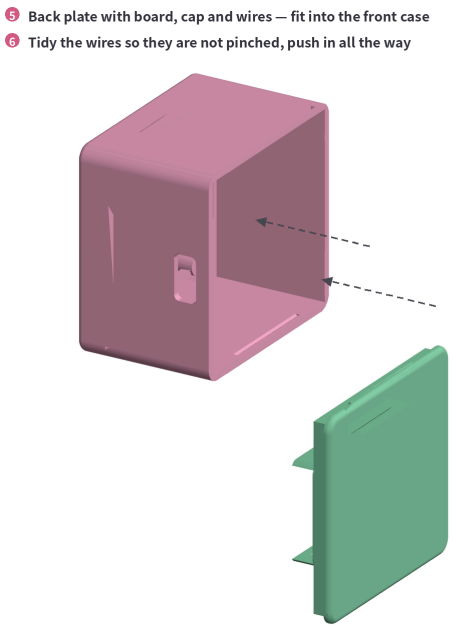
5-6. Plug in the jumper wires

❹ Jumper wires — plug into the board pins through the slots



1. Plug the other ends of the jumper wires from the display and the button module into the board pins through the slots of the cap.
2. Follow the tables and diagrams in chapter 2 for which wire goes to which pin.
3. Check the wire colors and pin numbers once more.

5-7. Close the case



1. Fit the back plate (with the board, cap and wires assembled) into the front case. Line it up so the USB-C port is on the same side as the opening in the side of the case.
2. Tidy the jumper wires so they are not folded or pinched, and push the plate in all the way.
3. Plug in a USB-C cable and check once more that the device powers on.

> If the back plate is too tight, do not force the edges. First check that no wire is pinched. To open it again, a thin plastic spudger can be used to widen the gap safely.